Exercise 3: SQL CASE Statements

Question 1

```
SELECT product_name,
       price,
   CASE
       WHEN price ≥ 1000 THEN 'Expensive'
       WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'
       WHEN price < 100 THEN 'Budget'
   END AS price_category
FROM products;
```

product_name	price	price_category
Laptop	1200	Expensive
Phone	800.00	Mid-range
Keyboard	45.00	Budget
Monitor	300.00	Mid-range
Mouse	25.00	Budget

Question 2

```
SELECT customer_name,
       amount,
   CASE
       WHEN amount ≥ 1000 THEN 'High Value'
       WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium Value'
       WHEN amount < 500 THEN 'Low Value'
   END AS order_value_category
FROM orders;
```

Customer_name	amount	order_value_category
Alice	150.0	Low Value
Bob	560.00	Medium Value
Charlie	999.99	Medium Value
Diana	45.50	Low Value
Ethan	1200.00	High Value

Question 3

```
SELECT emp_name,
       department,
       salary,
  CASE
     WHEN department = 'IT' AND salary > 80000 THEN 'Senior IT'
     WHEN department = 'HR' AND salary > 55000 THEN 'Experienced HR'
     ELSE 'Staff'
  END AS position_level
FROM employees;
```

emp_name	department	Salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

Question 4

```
SELECT student_name,
       score,
   CASE
      WHEN score ≥ 90 THEN 'A'
      WHEN score BETWEEN 80 AND 89 THEN 'B'
      WHEN score BETWEEN 70 AND 79 THEN 'C'
      WHEN score BETWEEN 60 AND 69 THEN 'D'
      WHEN score < 60 THEN 'F'
  END AS grade
FROM students;
```

student_name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

Question 5

```
SELECT delivery_id,
       delivery_time_minutes,
    CASE
      WHEN delivery_time_minutes ≤ 30 THEN 'Fast'
      WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN 'ON Time'
      ELSE 'Late'
   END AS performance
FROM deliveries;
```

delivery_id	delivery_time_minutes	performance
1	45	On Time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

Question 6

```
SELECT issue_type,
       priority,
    CASE
      WHEN priority = 3 THEN 'High'
      WHEN priority = 2 THEN 'Medium'
      WHEN priority = 1 THEN 'Low'
  END AS priority_label
FROM tickets;
```

issue_type	priority	priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

Question 7

```
SELECT student_id,
       (days_present/total_days) x100 AS attendance_percentage,
    CASE
      WHEN attendance_percentage ≥ 90 THEN 'Excellent'
      WHEN attendance_percentage BETWEEN 75 AND 89 THEN 'Good'
      WHEN attendance_percentage < 75 THEN 'Needs Improvement'
  END AS attendance_status
  FROM attendance;
```

student_id	attendance_percentage	attendance_status
1	90	Excellent
2	60	Needs Improvement
3	96	Excellent
4	50	Needs Improvement
5	100	Excellent

Question 8

```
SELECT product_id,
       stock_qty,
    CASE
      WHEN stock_qty =0 THEN 'Out of Stock'
      WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'
      ELSE 'In Stock'
  END AS stock_status.
  FROM products_inventory;
```

product_id	stock_qty	stock_status
1	5	Low Stock
2	0	Out of Stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

Question 9

```
SELECT subject,
       enrolled_students,
  CASE
    WHEN enrolled_students ≥ 25 THEN 'Large'
    WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
    WHEN enrolled_students <10 THEN 'Small'
 END AS class_size_category
 FROM classes;
```

subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

Question 10

```
SELECT payment_id,
       payment_method,
       amount,
  CASE
    WHEN payment_method = 'Cash' AND amount ≥ 200 THEN 'Eligible for Discount'
    ELSE 'Not Eligible'
 END AS discount_eligibility
FROM payments;
```

payment_id	payment_method	amount	discount_eligibility
1	Card	50.00	Not Eligible
2	Cash	200.00	Eligible for Discount
3	Card	150.00	Not Eligible
4	PayPal	75.00	Not Eligible
5	Cash	300.00	Eligible for Discount